

Q1 Nested For each and Data cleaning

Pig Script

-- Load data

```
students = LOAD 'data'
```

```
AS (name (chararray), grades: bag {t: (score  
:prop)}
```

-- Process: each student

```
result = FOREACH students {
```

block = remove 'INC' entries

```
clean-grades = FILTER grades BY score != 'INC'
```

Convert to integer

```
(int-grades) = FOREACH clean-grades
```

```
GENERATE (int)score AS score;
```

-- calculate average.

```
GENERATE
```

```
name,
```

```
AVG (int-grades.score) AS avg-score;
```

```
};
```

-- output

```
DUMP result;
```


Q2 Parameter Substitution and multistorage

Pig Script

```
-- load dataset.
sales = LOAD 'sales.csv'
      USING Pigstorage(',')
      AS (OrderID:int, Category:chararray,
          Amount:int);

-- split data using threshold.
SPLIT sales INTO
    high-value IF Amount > $threshold,
    standard-value IF Amount <= $threshold;

-- store results
STORE high-value INTO 'output/high-value'
    USING Pigstorage(',');
STORE standard-value INTO 'output/standard-
    value' USING
    Pigstorage(',');
```

Running → pig -param threshold=1000 sales.pig

Q3 Schema design & Versioning.

Create table

```
CREATE 'StockQuotes', { Name => 'price-da
                        Versions => 3 }
```


2) Insert 4 price updates.

```
put 'stockQuotes', 'AAPL', 'price-data:price',  
    '150'  
put 'stockQuotes', 'AAPL', 'price-data:price',  
    '155'  
put 'stockQuotes', 'AAPL', 'price-data:price', '160'  
put 'stockQuotes', 'AAPL', 'price-data:price', '165'
```

3) Retrieve all versions

```
get 'stockQuotes', 'AAPL', {COLUMN => 'price-data:  
    price', VERSIONS => 3}
```

Q4 Filters and Rowkey Scanning.

Scan 'UserActivity', {

 FILTER => "prefixFilter('user123') AND

 ValueFilter(=, 'binary:LOGIN')"

}

Q5 Atomic Counters & Increments

1) Create table

→ create 'Inventory', 'stats'

2) Initialize click-count = 0
put 'Inventory', 'sku-99', 'stock-count', 10

3) Atomic Increment

incr 'Inventory', 'sku-99', 'stock-count', 1

get 'Inventory', 'sku-99', 'stock-count'

print 'stock-count' = 11

wait 1000ms

get 'Inventory', 'sku-99', 'stock-count'